Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from lifelines import WeibullAFTFitter, LogNormalAFTFitter, LogLognormalAFTFitter
```

Data & Preprocessing

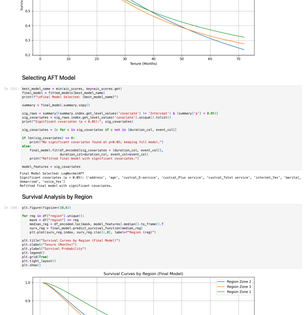
```
df = pd.read_csv("churn.csv")

df = df.drop(columns=["Surname", "CharmFl"])
df = df[df["Gender"] != 0].copy()

df["Gender"] = df["Gender"].map({"M": 1, "Ms": 0})

df["CharmFl"] = df.eq(-9999999999).any(axis=1).astype(int)
df = df.drop(columns=["CharmFl"])

IP_TOP = 20 # options:
df["IP"] = df["IP"].fillna("UNKNOWN")
df["IP"] = df["IP"].astype("category")

ip_counts = df["IP"].value_counts()
ip_keep = ip_counts[ip_counts >= IP_TOP].index.tolist()

ohe_cols = df["IP"].isin(ip_keep).astype(int).rename("IP_Top20")
df_encoded = pd.get_dummies(df, columns=["IP"], drop_first=True)

duration_col = "Tenure"
event_col = "Churn"
```

Fitting AFT Models

```
models = {
    "WeibullAFT": WeibullAFTFitter(),
    "LogLognormalAFT": LogLognormalAFTFitter(),
    "LogNormalAFT": LogNormalAFTFitter()
}

min_score = 0.1
min_pvalue = 0.05

for name, model in models.items():
    model.fit(df_encoded, duration_col=duration_col, event_col=event_col)
    print(f"\n{name} Summary:")
    if hasattr(model, "summary"):
        print(model.summary.head())
    else:
        print(model.print_summary())

scores = {}
for name, model in models.items():
    cindex = model.concordance_index_
    scores[name] = cindex

WeibullAFT Summary:
                                      coef  exp(coef)  se(coef)  coef lower 95%  \
param     covariate
lambda_   address                  0.04461    1.04562   0.00601         0.03285
          age                      0.02963    1.03010   0.00748         0.01497
          contract_0 service       0.07150    1.07413   0.00190         0.07149
          contract_1 service       0.70767    2.03544   0.01365         0.70742
          contract_Total service   0.88668    2.42946   0.01755         0.88624
          est_25k2 undergraduate   0.41675    1.51602   0.04276         0.33299
          est_25k3 undergraduate   0.39969    1.49151   0.04101         0.31927
          est_25k4 undergraduate   0.42353    1.52655   0.04347         0.33812
          est_home college         0.39969    1.49151   0.04101         0.31927
          gender_Male              0.04103    1.04189   0.01055         0.01993
          income                   0.00105    1.00105   0.00026         0.00055
          interest_free            0.17063    0.84155   0.01434         -0.14397
          marital_Unmarried        0.08898    1.09303   0.00427         0.00100
          region_Zone 2            0.08276    0.92075   0.01665         0.05020
          region_Zone 3            0.07449    1.07278   0.01757         0.03938
          retire_free              0.70566    2.02437   0.02287         0.65835
          retire_Yes               0.55787    0.71937   0.18444         -0.82453
          intercept                1.79715    6.03197   0.01557         1.69355
          interrupt                0.74663    2.10610   0.02073         0.74004
                                   coef upper 95%  exp(coef) lower 95%  \
param     covariate
lambda_   address                 0.05637         1.03346     1.06577
          age                     0.04429         1.01502     1.05127
          contract_0 service      0.07151         1.07411     1.07415
          contract_1 service      0.70792         2.03491     2.03597
          contract_Total service  0.88712         2.42860     2.43032
          est_25k2 undergraduate  0.50051         1.39648     1.64666
          est_25k3 undergraduate  0.48012         1.37500     1.61838
          est_25k4 undergraduate  0.51894         1.40800     1.68000
          est_home college        0.48012         1.37500     1.61838
          gender_Male             0.06213         1.01960     1.06411
          income                  0.00155         1.00055     1.00155
          interest_free           0.44269         0.65800     1.55811
          marital_Unmarried       0.16696         1.00101     1.18151
          region_Zone 2           0.11532         0.89312     0.95115
          region_Zone 3           0.10960         1.03946     1.11548
          retire_free             0.75297         1.93197     2.12313
          retire_Yes              0.81121         0.43811     2.25000
          intercept               1.90076         5.42800     6.69151
          interrupt               0.79322         2.07073     2.21024
                                  exp(coef) upper 95%  se(coef) lower 95%  \
param     covariate
lambda_   address                   1.07566     0.00604         0.03245
          age                       1.03046     0.00751         0.01509
          contract_0 service        1.07418     0.00190         0.07146
          contract_1 service        2.03803     0.01365         0.70715
          contract_Total service    2.43391     0.01755         0.88611
          est_25k2 undergraduate    1.65148     0.04276         0.41642
          est_25k3 undergraduate    1.63000     0.04101         0.39916
          est_25k4 undergraduate    1.68200     0.04347         0.42329
          est_home college          1.63000     0.04101         0.39916
          gender_Male               1.06396     0.01055         0.04093
          income                    1.00156     0.00026         0.00104
          interest_free             1.56270     0.01434         -0.14371
          marital_Unmarried         1.18251     0.00427         0.08893
          region_Zone 2             0.95315     0.01665         0.08275
          region_Zone 3             1.11553     0.01757         0.07446
          retire_free               2.12371     0.02287         0.70564
          retire_Yes                2.25000     0.18444         -0.82453
          intercept                 6.69151     0.01557         1.79715
          interrupt                 2.21024     0.02073         0.74663
                                   coef upper 95%  exp(coef) lower 95%  \
param     covariate
rho_      address                 0.03663         1.03804     1.03785
          age                     0.02427         1.02457     1.02437
          contract_0 service      0.07154         1.07418     1.07415
          contract_1 service      0.70792         2.03491     2.03597
          contract_Total service  0.88726         2.42860     2.43032
          est_25k2 undergraduate  0.42335         0.97609     0.97579
          est_25k3 undergraduate  0.40026         0.97609     0.97579
          est_25k4 undergraduate  0.42878         0.97609     0.97579
          est_home college        0.40026         0.97609     0.97579
          gender_Male             0.04113         0.99960     0.99960
          income                  0.00106         1.00104     1.00104
          interest_free           0.44269         0.86631     0.86631
          marital_Unmarried       0.09707         0.99960     0.99960
          region_Zone 2           0.08277         0.99960     0.99960
          region_Zone 3           0.07452         0.99960     0.99960
          retire_free             0.70588         0.99960     0.99960
          retire_Yes              0.81121         0.99960     0.99960
          intercept               1.79715         0.99960     0.99960
          interrupt               0.74663         0.99960     0.99960
                                   exp(coef) upper 95%  se(coef) lower 95%  \
param     covariate
LogNormAFT Summary:
                                      coef  exp(coef)  se(coef)  coef lower 95%  \
param     covariate
alpha_    address                 0.03663    1.03804   0.00604         0.02495
          age                     0.02441    1.02469   0.00751         0.01000
          contract_0 service      0.07154    1.07418   0.00190         0.07153
          contract_1 service      0.70792    2.03491   0.01365         0.70767
          contract_Total service  0.88726    2.42860   0.01755         0.88691
          est_25k2 undergraduate  0.42335    1.52657   0.04347         0.33812
          est_25k3 undergraduate  0.40026    1.49151   0.04101         0.31927
          est_25k4 undergraduate  0.42878    1.53600   0.04347         0.34354
          est_home college        0.40026    1.49151   0.04101         0.31927
          gender_Male             0.04113    1.04201   0.01055         0.01993
          income                  0.00106    1.00106   0.00026         0.00055
          interest_free           0.44269    1.55811   0.01434         -0.14397
          marital_Unmarried       0.09707    1.10192   0.00427         0.00100
          region_Zone 2           0.08277    0.92054   0.01665         0.05020
          region_Zone 3           0.07452    1.07298   0.01757         0.03938
          retire_free             0.70588    2.02437   0.02287         0.65835
          retire_Yes              0.81121    0.43811   0.18444         -0.82453
          intercept               1.79715    6.03197   0.01557         1.69355
          interrupt               0.74663    2.10610   0.02073         0.74004
                                   coef upper 95%  exp(coef) lower 95%  \
param     covariate
sigma_    address                 0.04461         1.04562     1.04461
          age                     0.02963         1.03010     1.02963
          contract_0 service      0.07150         1.07413     1.07411
          contract_1 service      0.70767         2.03544     2.03491
          contract_Total service  0.88668         2.42946     2.42860
          est_25k2 undergraduate  0.50854         1.66000     1.52657
          est_25k3 undergraduate  0.48012         1.61838     1.49151
          est_25k4 undergraduate  0.51371         1.67000     1.53600
          est_home college        0.48012         1.61838     1.49151
          gender_Male             0.06213         1.06412     1.04201
          income                  0.00157         1.00157     1.00106
          interest_free           0.44269         0.86631     1.55811
          marital_Unmarried       0.18693         1.20000     1.10192
          region_Zone 2           0.11532         0.89312     0.92054
          region_Zone 3           0.10960         1.06412     1.07298
          retire_free             0.75297         1.93197     2.02437
          retire_Yes              0.81121         0.43811     0.43811
          intercept               1.90076         5.42800     6.03197
          interrupt               0.79322         2.07073     2.10610

LogLognormAFT Summary:
                                      coef  exp(coef)  se(coef)  coef lower 95%  \
param     covariate
mu_       address                 0.03663    1.03804   0.00604         0.02495
          age                     0.02441    1.02469   0.00751         0.01000
          contract_0 service      0.07154    1.07418   0.00190         0.07153
          contract_1 service      0.70792    2.03491   0.01365         0.70767
          contract_Total service  0.88726    2.42860   0.01755         0.88691
          est_25k2 undergraduate  0.42335    1.52657   0.04347         0.33812
          est_25k3 undergraduate  0.40026    1.49151   0.04101         0.31927
          est_25k4 undergraduate  0.42878    1.53600   0.04347         0.34354
          est_home college        0.40026    1.49151   0.04101         0.31927
          gender_Male             0.04113    1.04201   0.01055         0.01993
          income                  0.00106    1.00106   0.00026         0.00055
          interest_free           0.44269    1.55811   0.01434         -0.14397
          marital_Unmarried       0.09707    1.10192   0.00427         0.00100
          region_Zone 2           0.08277    0.92054   0.01665         0.05020
          region_Zone 3           0.07452    1.07298   0.01757         0.03938
          retire_free             0.70588    2.02437   0.02287         0.65835
          retire_Yes              0.81121    0.43811   0.18444         -0.82453
          intercept               1.79715    6.03197   0.01557         1.69355
          interrupt               0.74663    2.10610   0.02073         0.74004
                                   coef upper 95%  exp(coef) lower 95%  \
param     covariate
sigma_    address                 0.04461         1.04562     1.04461
          age                     0.02963         1.03010     1.02963
          contract_0 service      0.07150         1.07413     1.07411
          contract_1 service      0.70767         2.03544     2.03491
          contract_Total service  0.88668         2.42946     2.42860
          est_25k2 undergraduate  0.50854         1.66000     1.52657
          est_25k3 undergraduate  0.48012         1.61838     1.49151
          est_25k4 undergraduate  0.51371         1.67000     1.53600
          est_home college        0.48012         1.61838     1.49151
          gender_Male             0.06213         1.06412     1.04201
          income                  0.00157         1.00157     1.00106
          interest_free           0.44269         0.86631     1.55811
          marital_Unmarried       0.18693         1.20000     1.10192
          region_Zone 2           0.11532         0.89312     0.92054
          region_Zone 3           0.10960         1.06412     1.07298
          retire_free             0.75297         1.93197     2.02437
          retire_Yes              0.81121         0.43811     0.43811
          intercept               1.90076         5.42800     6.03197
          interrupt               0.79322         2.07073     2.10610

AIC Scores:
WeibullAFT: 2904.3452
LogLognormAFT: 2916.8149
LogNormAFT: 2919.8149
```

Survival Curves

```
feature_columns = [c for c in df_encoded.columns if c not in [duration_col, event_col]]
median_customer = df_encoded[feature_columns].median().to_frame().T

plt.figure(figsize=(8,6))
for name, model in models.items():
    sf = model.predict_survival_function(median_customer)
    plt.plot(sf.index, sf[sf.columns[0]], label=name)

plt.title("AFT Survival Curves (Median Customer)")
plt.xlabel("Tenure (Months)")
plt.ylabel("Survival Probability")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

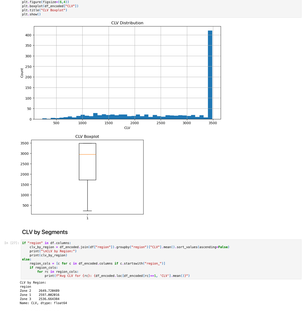


Selecting AFT Model

```
best_model_name = scores.keys().max()
best_model = models[best_model_name]
print("Selected Model Selected: {best_model_name}")

summary = best_model.summary.copy()

sig_rows = summary[summary.index.get_level_values('covariate') != 'Intercept'] & (summary['p'] < 0.05)]
sig_covariates = sig_rows.index.get_level_values('covariate').unique().tolist()
print("Significant Covariates (p < 0.05): {len(sig_covariates)}")

sig_covariates = [c for c in sig_covariates if c not in [duration_col, event_col]]

if len(sig_covariates) == 0:
    raise ValueError("No covariates found at p<0.05; keeping full model.")

cox_df = df_encoded[
    df_encoded[duration_col].notnull() & df_encoded[event_col].notnull() &
    (df_encoded[duration_col] > 0) & df_encoded[event_col].isin([0,1])
].copy[[duration_col, event_col, *sig_covariates]]

print("Retained {len(cox_df.columns)-2} model features")
```

Survival Analysis by Region

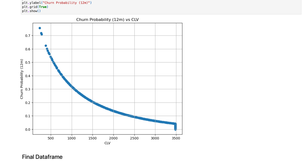
```
plt.figure(figsize=(10,6))

for reg in df["region"].unique():
    print("Processing {reg}...")
    median_reg = df_encoded.loc[mask, model_features].median().to_frame().T
    mask = (df["region"] == reg) & df_encoded[duration_col].notnull()
    pf = best_model.predict_survival_function(median_reg, times=sorted(df[duration_col].unique()))
    plt.plot(pf.index, pf[pf.columns[0]], label=f"Region {reg}")

plt.title("Survival Curves by Region (Final Model)")
plt.xlabel("Tenure (Months)")
plt.ylabel("Survival Probability")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



Predicted Lifetime and CLV

```
predicted_lifetime = final_model.predict_median(df_encoded[model_features])
df["predicted_lifetime"] = predicted_lifetime.values

monthly_interest = 0.02
discount_rate = 0.05

df_encoded["predicted_lifetime_capped"] = df_encoded["predicted_lifetime"].clip(lower=0.5, upper=120)

t = df_encoded["predicted_lifetime_capped"]
df_encoded["CLV"] = (monthly_revenue * (t - 1)) / (1 + discount_rate) ** (t-0.5)
```

```
plt.figure(figsize=(10,5))
plt.hist(df["CLV"], bins=40)
plt.xlabel("CLV")
plt.ylabel("Count")
plt.title("CLV Distribution")
plt.show()

plt.figure(figsize=(8,4))
plt.boxplot(df_encoded["CLV"])
plt.ylabel("CLV Boxplot")
plt.show()
```



CLV by Segments

```
df["region"] = df["region"]
df["contract"] = df["contract"]
cls_by_region = df_encoded.loc[df["region"].].groupby("region")["CLV"].mean().sort_values(ascending=False)
print("Average CLV by Region")
print(cls_by_region)

for contract in df["contract"].unique:
    cls = df_encoded.loc[df["contract"] == contract]["CLV"]
    print("{contract}: {cls.mean()}")
    for region_zone:
        mask = (df["region"] == region_zone) & (df["contract"] == contract)
        if mask.sum() > 0:
            print("  {region_zone}: {df_encoded.loc[df_encoded[mask].CLV].mean()}")

CLV by Region
region
Zone 1    2645.726600
Zone 3    2541.484864
Zone 2    2511.000000
Name: CLV, dtype: float64

CLV by Region
cls_by_region.plot(kind="bar", figsize=(8,5))
plt.title("Average CLV by Region")
plt.ylabel("Average CLV")
plt.xticks(rotation=0)
plt.show()

for contract in df["contract"]:
    cls_by_contract = df_encoded.loc[df["contract"]=="{contract}"].mean()
    cls_by_contract.plot(kind="bar", figsize=(8,5))
    plt.title("Average CLV by Customer Category")
    plt.ylabel("Average CLV")
    plt.xticks(rotation=0)
    plt.show()
```


Estimated Retention Budget

```
churn_12 = final_model.predict_survival_function(df_encoded[model_features], times=[12])
df["churn_prob_12"] = churn_12.values
churn_prob_12 = churn_12.values

expected_budget = (churn_prob_12 * df_encoded["CLV"].values).sum()
print("Estimated Annual Retention Budget: {expected_budget:,.2f}")

Suggested Annual Retention Budget: 327884.30
```

Churn vs CLV

```
plt.figure(figsize=(8,6))
plt.scatter(df_encoded["CLV"], churn_prob_12)
plt.xlabel("Customer Predictability Index")
plt.ylabel("Churn Probability")
plt.title("Churn Probability (12m) vs Customer Predictability Index")
plt.grid(True)
plt.show()
```


Final Dataframe

```
df = df_encoded.copy()
df["CLV"] = df_encoded.
df.to_csv("churn_predictions_with_clv.csv", index=False)
```